Solutions to kilpamatopas

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1 Solutions

Exercise 1.1. Find b such that $1988x^2 + bx + 8891$ and $8891x^2 + bx + 1988$ share a root.

Solution to exercise 1.1. When they share a root, this equation holds:

$$1988x^2 + bx + 8891 = 8891x^2 + bx + 1988$$
$$(1988 - 8891)x^2 + (8891 - 1988) = 0$$
$$-6903x^2 + 6903 = 0$$
$$6903x^2 = 6903$$
$$x^2 = 1$$

The common root must be -1 or 1. Then using Vieta's formulas on the first polynomial:

$$\pm r_2 = \frac{c}{a}$$
$$r_2 = \pm\frac{c}{a}$$
$$\pm 1 + r_2 = -\frac{b}{a}$$
$$\pm(1 + \frac{c}{a}) = -\frac{b}{a}$$
$$\pm(a + c) = b$$
$$b = \pm 10879$$

□

Exercise 1.2. Let $a, b, c, d \in \mathbb{R}^+$. Prove that

$$\frac{1}{a} + \frac{1}{b} + \frac{4}{c} + \frac{16}{d} \geq \frac{64}{a + b + c + d}$$

Solution to exercise 1.2. We can write the LHS as:

$$\frac{1^2}{a} + \frac{1^2}{b} + \frac{2^2}{c} + \frac{4^2}{d}$$

Cauchy-Schwartz in Engel form states that:

$$\frac{(\sum_{k=1}^{n} x_k)^2}{\sum_{k=1}^{n} y_k} \leq \sum_{k=1}^{n} \frac{x_k^2}{y_k}$$

Using Cauchy-Schwartz with: $x_1 = (1, 1, 2, 4)$ and $y_1 = (a, b, c, d)$, we get:

$$\frac{8^2}{a+b+c+d} \leq \frac{1}{a} + \frac{1}{b} + \frac{4}{c} + \frac{16}{d}$$
$$\frac{64}{a+b+c+d} \leq \frac{1}{a} + \frac{1}{b} + \frac{4}{c} + \frac{16}{d}$$

□